

Enhancing YouTube Data Donations With Platform Metadata

Anonymized website companion for review

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This PDF is an anonymized review copy of the public companion website. It summarizes the website’s purpose, workflow, mock data, legal/ethical notes, and notebook guide without repository or authorship-identifying sections.

This repository demonstrates how donated YouTube account data can be linked with video- and channel-level platform metadata to create **enriched exposure histories**.

The project accompanies the method described in *Enhancing YouTube Data Donations With Platform Metadata*. It shows, in reproducible form, how researchers can move from raw Google Takeout-style files, as obtained via data donations, to event-level indicators of YouTube exposure: what was watched, when it was watched, estimated watch time, whether it followed a search, whether it came from a subscribed channel, whether it was a Short or Long, and whether later metadata collection suggests that a video became unavailable for potentially problematic content-related reasons. For demonstration purposes this repository uses **synthetic mock data**.

Purpose

YouTube watch histories are valuable for media, journalism and platform research because they record concrete viewing behavior over time rather than recall and report biased self-reports, aggregate platform metrics, or metadata by themselves. On their own, however, donated watch histories are sparse.

This project shows how those traces can be enriched by joining them with video-level metadata through the YouTube `video_id`. After enrichment, each watch event can be analyzed not only as a timestamped view, but as part of a wider exposure history containing information about video format, channel context, likely viewing pathway, availability status, and time of day. In aggregate this allows inference not only on when

but also on how data donors used the platform and provides additional information on what the people encountered.

The workflow is meant for researchers who want to understand or adapt a data donation pipeline for YouTube-related research, especially in journalism studies, platform accountability research, computational social science, and media-use research.

What This Repository Demonstrates

The repository shows how to:

- clean donated YouTube watch, search, and subscription files;
- extract and standardize YouTube `video_ids` across donated files;
- prepare a video-level metadata table;
- join donated histories and metadata at the event level;
- classify watches as search-associated or subscription-linked where possible;
- distinguish Shorts from Longs using metadata-derived rules;
- classify viewing as daytime or nighttime according to a configurable time window;
- estimate watch time from metadata duration and the next observed watch event;
- attach availability and problematic-view indicators based on metadata/error signals;

The goal is not to reconstruct every individual viewing decision. YouTube does not provide authoritative explanations for why a specific video appeared in a user's history and we have no access to what a given video meant to a given viewer at time of viewing. However, the workflow produces **aggregate exposure indicators** that allow insight into how YouTube use differs across users.

What Enrichment Adds

A raw watch-history row tells us that a participant started watching a video at a particular time. After enrichment, the same row can also include:

- video duration;
- channel and channel identifier;
- upload date;
- estimated watch time;
- view, like, and comment metadata;
- language, format, aspect ratio, and other video-level fields;
- whether the video is classified as a Short or Long;
- whether the watch happened during the configured day or night period;
- indication of whether the watch followed an explicit search;
- indication of whether the video came from a channel the user is subscribed to;
- whether a watched video was later removed by YouTube;
- whether the removal was due to potentially problematic content of the video

These data can then be used for user-level assessments like:

- share of searched or subscribed watches among all watches
- estimated watch time across participants, formats, and routines
- distribution of Shorts/Longs across day/night watches
- frequency of encountering potentially problematic content

These indicators are imperfect but useful. They are strongest when used to describe patterns across many events or participants, rather than to make definitive claims about single views. The figures below exemplify how the data can be used to quantify YouTube usage and identify differences between individual data donors. These examples are illustrative outputs from the synthetic mock data.

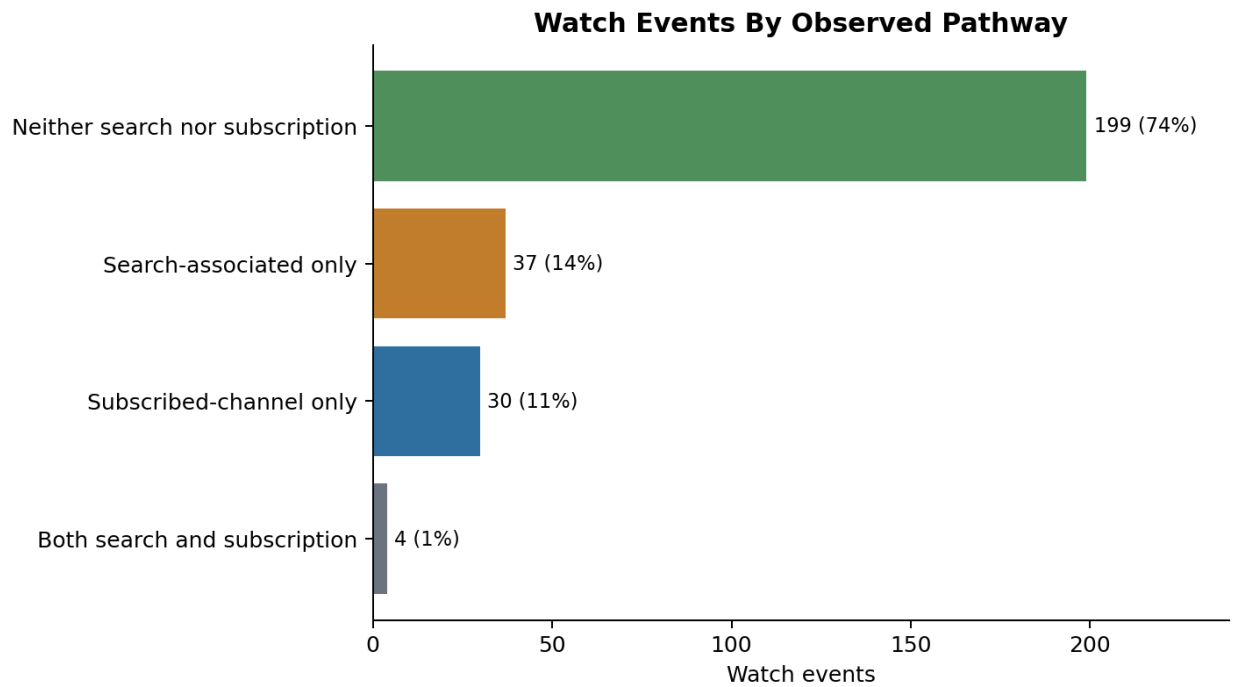


Figure 1: Search and subscription context make it possible to distinguish whether watches followed explicit searches, came from subscribed channels, both, or neither.

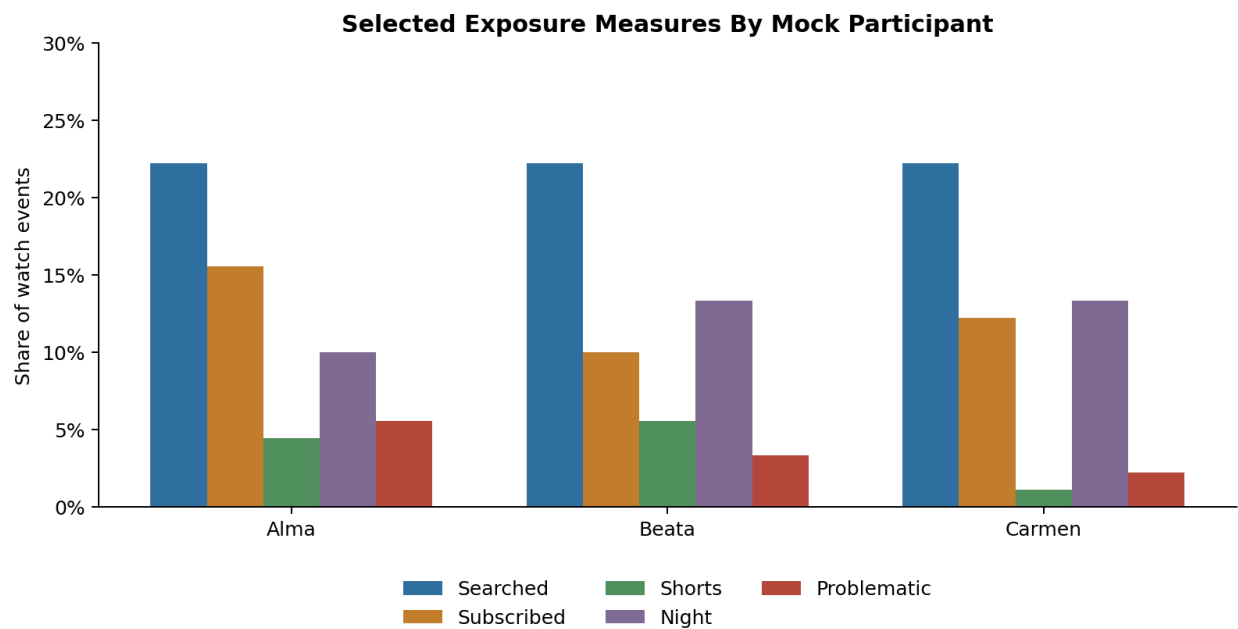


Figure 2: The enriched event table allows participant-level comparison across searched, subscribed, Shorts, nighttime, and problematic-view indicators.

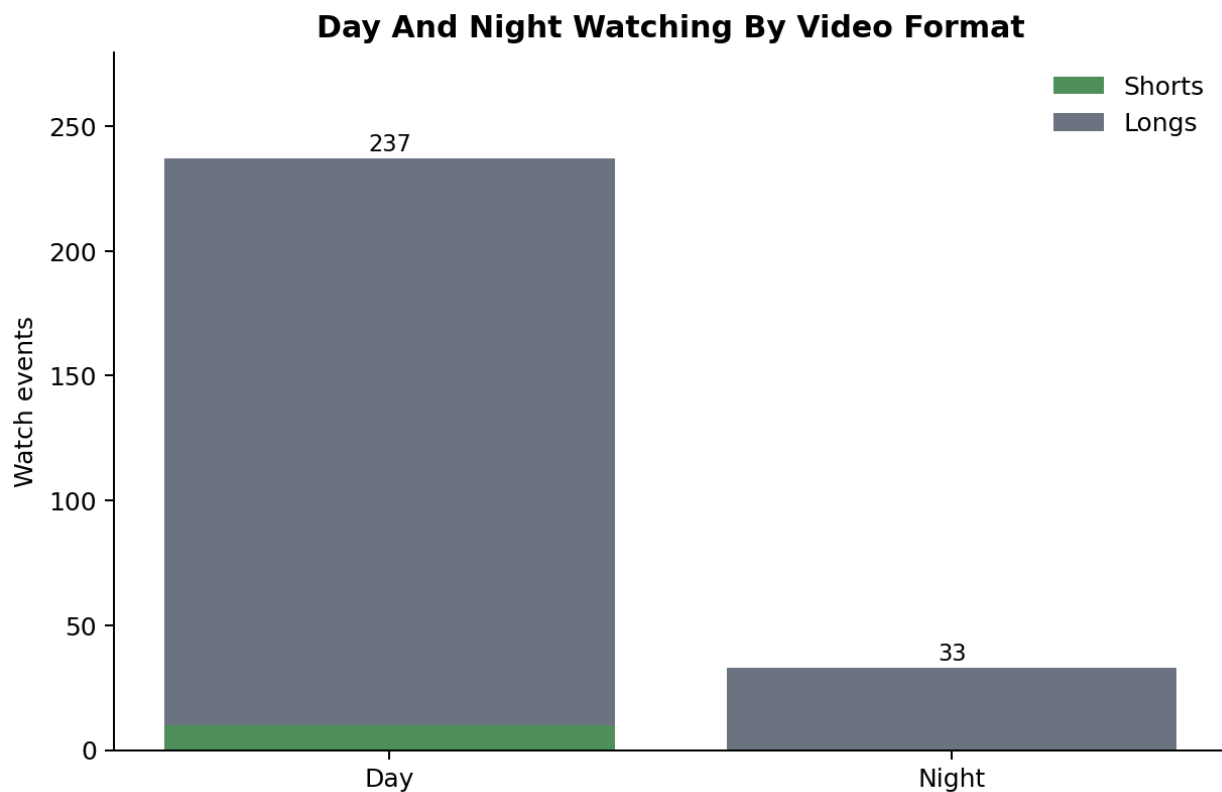


Figure 3: Metadata-derived format labels and localized timestamps allow analyses of Shorts/Longs across day and night watching.

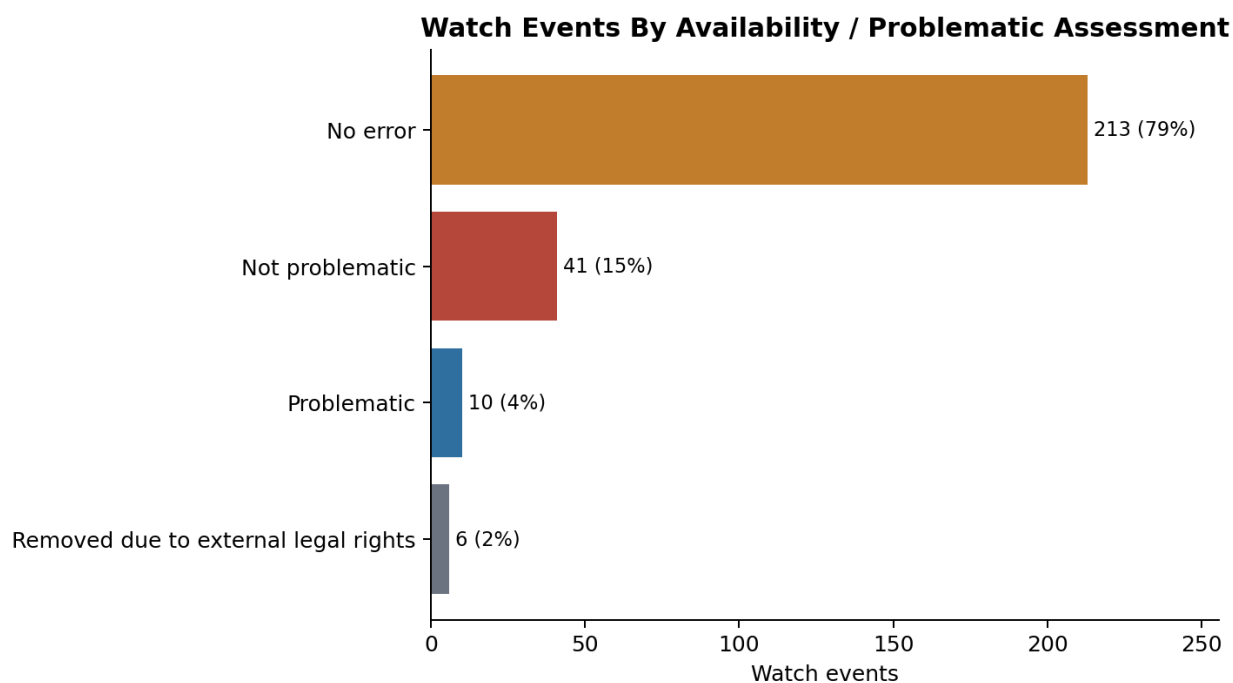


Figure 4: Scraped metadata errors and manual classifications make unavailable or potentially problematic watched videos analyzable at the event level.

Workflow

```
{mermaid} flowchart TD
    A["Google Takeout-data<br/>watch histories, search histories, subscriptions"] --> B["Clean donated histories"]
    C["Video-level metadata<br/>scraped metadata"] --> D["Prepare metadata table"]
    B --> E["Join by YouTube video_id"]
    D --> E
    E --> F["Event-level enriched watch history"]
    F --> G["Estimate watch time<br/>from duration and next observed watch"]
    G --> H["Compute exposure indicators<br/>search, subscriptions, Shorts/Longs, day/night, availability"]
    H --> I["Tables and figures"]
```

Recommended Entry Point

For a quick overview of the finished workflow, start with:

[Notebook 06: Demonstrate enriched exposure measures](#)

This notebook shows the final enriched data structure and produces compact figures that illustrate what the method makes visible.

Then use the [notebook guide](#) to inspect the earlier steps in order: cleaning donated histories, preparing metadata, demonstrating scraping logic, linking records, estimating watch time, and producing the final exposure measures.

Mock Data

The mock dataset contains three fictional donors: Alma, Beata, Carmen. Each donor folder contains small watch-history, search-history, and subscription files with the same broad structure as a YouTube Takeout export. The public identifiers and text fields are synthetic. Video IDs, titles, channel IDs, channel names, search queries, tags, and URLs have been replaced with deterministic fake values.

Legalities of Metadata Collection

[Notebook 03](#) demonstrates the metadata collection logic used in the method. In a real research project, metadata collection requires careful legal, ethical, and institutional assessment. It may also depend on the researcher's access rights, data security setup, and platform access conditions.

One route to obtain research access to YouTube is the Digital Services Act Article 40(12) for research with a systemic risk focus. Access can be requested via this [request records form](#) and requires provision of the scraper's IP and a *product token string* for authentication. Researchers with access can make themselves identifiable to YouTube by using their product token string as `IDENTIFYING_USER_AGENT` when scraping.

An example of a successful data access request can be found [here](#).

For further support in requesting data access under Article 40 of the Digital Services Act (DSA), the [DSA Collaboratory](#) is highly recommended.

Recommended Reading Paths

There are two useful ways to read the notebooks.

If you want the **quickest sense of what the method produces**, start with [Notebook 06](#). It shows the enriched event table and the final demonstration figures.

If you want to understand the **full workflow**, read the notebooks in order:

1. clean donated watch histories;
2. clean donated search histories;
3. document metadata collection;
4. link watch events, metadata, subscriptions, searches, and availability signals;

5. estimate watch time from metadata duration and the next observed watch event;
6. summarize the enriched exposure measures.

Pipeline Overview

```
{mermaid} flowchart TD
  A["01 Clean donated watch histories"] --> B["video_histories.csv"]
  C["02 Clean donated search histories"] --> D["search_query_histories.csv"]
  B --> E["03 Scrape or provide video metadata"]
  E --> F["meta_data.csv or public metadata fixture"]
  F --> G["04 Link watch events, metadata, and exposure context"]
  D --> G
  G --> H["Donated subscriptions"]
  H --> I["Manual availability/error classification"]
  I --> J["video_histories_enriched.csv"]
  J --> K["05 Estimate watch time"]
  K --> L["video_histories_watchtime.csv"]
  L --> M["06 Demonstrate enriched exposure measures"]
  M --> N["Demo tables and figures"]
```

Notebook Summary

Notebook	Main role	Main output
01 Clean Donated Watch Data	Turns raw Takeout-style watch histories into validated video-watch events.	outputs/tables/video_histories.csv
02 Clean Donated Search Data	Turns raw Takeout-style search histories into genuine YouTube search-query events.	outputs/tables/search_query_histories.csv
03 Scrape YouTube Metadata	Documents how video-level metadata can be collected with yt_dlp.	outputs/tables/meta_data.csv
04 Link Watch Data, Metadata, And Exposure Context	Joins watch events with metadata, search, subscription, time-of-day, format, and availability indicators.	outputs/tables/video_histories_enriched.csv
05 Estimate Watch Time	Estimates event-level watch time from metadata duration and the next observed watch event.	outputs/tables/video_histories_watchtime.csv
06 Demonstrate Enriched Exposure Measures	Shows what the enriched data makes visible and produces demonstration figures.	Demo tables and figures in outputs/

01 Clean Donated Watch Data

[Open notebook 01](#)

This notebook is the first cleaning step. It reads mock Google Takeout-style YouTube watch-history files from `data/mock_takeout`, combines donor files, standardizes columns, parses timestamps, extracts YouTube `video_ids`, and separates validated video watches from ads, YouTube Music rows, and other non-video rows.

It also cleans multilingual watch-action prefixes so that titles can be interpreted more consistently across Takeout exports.

The main handoff is:

- `outputs/tables/video_histories.csv`

This table keeps one row per retained YouTube video watch event. It contains the participant ID, timestamp, watched title, cleaned title, URL, extracted `video_id`, and channel information where available.

The notebook also writes side tables for auditing:

- `outputs/tables/watch_histories.csv`
- `outputs/tables/music_histories.csv`
- `outputs/tables/ad_watch_histories.csv`
- `outputs/tables/other_histories.csv`
- `outputs/tables/title_prefix_diagnostics.csv`

These side tables are useful because they show what was retained, what was excluded, and which localized title prefixes were recognized.

02 Clean Donated Search Data

[Open notebook 02](#)

This notebook cleans donated YouTube search-history files. It reads mock Takeout-style search histories, combines donor files, standardizes columns, parses timestamps, decodes search queries from URLs, and identifies rows that represent genuine YouTube searches.

The main handoff is:

- `outputs/tables/search_query_histories.csv`

This table is used later to determine whether a watch event occurred shortly after a participant searched on YouTube.

The notebook also writes:

- `outputs/tables/search_histories.csv`

The distinction matters because not every row in a Takeout search-history file is necessarily useful for the exposure pathway indicator. The later workflow relies only on rows classified as genuine search-query events.

03 Scrape YouTube Metadata

[Open notebook 03](#)

This notebook documents the metadata collection step. It reads unique `video_ids` from `outputs/tables/video_histories.csv` and shows how `yt_dlp` can be used to request public YouTube metadata for each video.

The intended output of a real metadata run is:

- `outputs/tables/meta_data.csv`

Each row represents one unique YouTube video ID. If metadata can be retrieved, the row contains video-level fields such as title, channel, duration, upload date, format information, popularity metrics, language, subtitle availability, and related metadata. If metadata cannot be retrieved, the row is still kept and the raw `yt_dlp` error message is stored.

This notebook is documented but not executed in the public site. The public repository contains scrambled fake video IDs, which preserve the shape of the workflow but cannot be scraped from YouTube.

To run this notebook in a real project, researchers would need authorized real YouTube IDs, appropriate legal and institutional permissions, and a registered identifying User-Agent token or equivalent project identifier. The notebook stops by default until `YOUR_REGISTERED_TOKEN` is replaced.

For public reproduction, later notebooks can use the scrambled fixture:

- `data/demo_metadata/meta_data_public.csv`

This allows readers to understand the linking workflow without running live YouTube metadata collection.

04 Link Watch Data, Metadata, And Exposure Context

[Open notebook 04](#)

This is the main enrichment notebook. It joins cleaned watch events to video-level metadata and adds the event-level indicators that turn raw watch histories into enriched exposure histories.

It reads:

- `outputs/tables/video_histories.csv`
- `outputs/tables/search_query_histories.csv`
- `outputs/tables/meta_data.csv`, if available
- `data/demo_metadata/meta_data_public.csv`, as the public fallback metadata fixture
- donated subscription files from `data/mock_takeout`
- `data/manual/unavailable_content_classification_filled.csv`

It writes the main enriched event table:

- `outputs/tables/video_histories_enriched.csv`

Each row is still one donated watch event, but now the event also carries metadata, linkage status, local-time helper fields, Shorts/Longs classification, day/night classification, search-associated markers, subscribed-channel markers, and availability/problematic-view labels.

The notebook also writes audit and helper files:

- `outputs/tables/metadata_linkage_report.csv`
- `outputs/tables/subscription_histories.csv`
- `outputs/tables/exposure_linkage_report.csv`
- `outputs/tables/error_types_by_video.csv`

Several methodological choices are made transparent in this notebook:

- the day/night boundary is configurable;
- the mock project uses **Europe/Copenhagen** as the default timezone;
- Shorts are classified from metadata using duration and aspect-ratio logic;
- search-associated views are marked when a participant search occurred 1 to 55 seconds before the watch event;
- subscription-linked views are identified by comparing metadata channel IDs with donated subscription channel IDs;
- raw metadata errors are preserved before being generalized and manually classified.

The notebook keeps the unit of analysis at the watch-event level. No donated watch event is dropped simply because metadata is missing or returned an error.

05 Estimate Watch Time

[Open notebook 05](#)

This notebook estimates watch time from the enriched event table created by notebook 04. It uses the original workflow's core idea: if a participant starts another video before the current video's metadata duration has elapsed, the remaining duration is treated as skipped.

It reads:

- `outputs/tables/video_histories_enriched.csv`

It writes:

- `outputs/tables/video_histories_watchtime.csv`
- `outputs/tables/watchtime_summary.csv`
- `outputs/tables/watchtime_diagnostics.csv`

The event-level output keeps one row per watch event and adds helper fields such as the next observed watch time, seconds until the next watch, estimated watch seconds, and a watch-time status flag. The summary collapses those estimates across the full mock dataset by participant, Shorts/Longs, and day/night.

This estimate should not be interpreted as exact viewing completion. It is a transparent approximation based on observed start times and video duration metadata.

06 Demonstrate Enriched Exposure Measures

[Open notebook 06](#)

This is the showcase notebook. It starts from the enriched event table produced by notebook 04 and demonstrates what the method makes visible.

It reads:

- `outputs/tables/video_histories_enriched.csv`
- `outputs/tables/video_histories_watchtime.csv`, when available
- `outputs/tables/exposure_linkage_report.csv`, when available

It writes:

- `outputs/tables/demo_enriched_event_examples.csv`
- `outputs/tables/demo_measure_summary.csv`
- `outputs/figures/demo_pathway_mix.png`
- `outputs/figures/demo_participant_measure_shares.png`
- `outputs/figures/demo_daynight_shorts.png`
- `outputs/figures/demo_problematic_views.png`

The notebook first displays selected enriched watch events. These examples show how a single donated watch event can now be read together with metadata fields, pathway indicators, format labels, day/night labels, and availability/problematic-view signals.

It then creates four compact figures:

1. **Pathway mix**
Shows how many watched events are search-associated, subscription-linked, both, or neither.
2. **Participant-level measure shares**
Summarizes exposure indicators by fictional donor in the mock dataset.
3. **Day/night and Shorts/Longs**
Combines local viewing time with metadata-derived format labels.
4. **Problematic and unavailable signals**
Demonstrates how later metadata errors and manual classification can be summarized as availability or problematic-view indicators.